# Biosignal Analysis – Math Fundamentals Test

## Questions

### Question Set 1: Matrix Multiplications

Write down the result of the following matrix multiplications:

$$[1 \quad 1] \cdot \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \ldots \tag{1.1}$$

$$\begin{bmatrix} 3 \\ 4 \end{bmatrix} . [1 \quad 1] = \ldots \tag{1.2}$$

$$\begin{bmatrix} 2 \\ 4 \end{bmatrix} . \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \ldots \tag{1.3}$$

### Question Set 2: Geometry

What is the line defined by the two points $\vec{x}_1 = [1 \quad 1]$, $\vec{x}_2 = [3 \quad 7]$, in the form

$$g: \; y = k * x + d \; ? \tag{2.1}$$

What is the unit vector (i.e. length =1) perpendicular to that line?

$$\vec{n} \perp g, with |\vec{n}| = 1 \tag{2.2}$$

What is the distance *d* of the point $\vec{x}_3 = [5 \quad 1]$ from that line? (2.3)

What is the vector perpendicular to $\vec{p} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \vec{q} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$ (2.4)

### Question Set 3: Complex Numbers

Sketch the real and the complex part of

$$e^{ix} \tag{3.1}$$

for $0 < x < 4 * \pi$.

Convert the following complex number into polar coordinates $(r, \theta)$

$$c = 3 + i * 1 \tag{3.2}$$